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SVKM'S NMIMS
Mukesh Patel School of Technology Management & Engineering
Department of Computer Engineering
Java Lab
Subject- Object Oriented Programming through JAVA

EXPERIMENT NO. 3

Aim:
To study and implement Object Oriented Programming concepts such as Class, Object, Methods, Constructor, and Polymorphism (Method Overloading) using Java.

Materials:
1. Editor – Notepad/Notepad++
2. Command Prompt

Theory:

- **Class**: A blueprint that defines properties and behaviors of an object.
- **Syntax**

```
class ClassName {
// variables
// methods
}
```

- **Object**: An instance of a class.
- **Syntax**

```
ClassName objectName = new ClassName();
```

- **Method**: A function defined inside a class to perform a specific task.
- Syntax

```
returnType methodName(parameters) {
// method body
}
```

- **Constructor**: A special method used to initialize objects.
- **Syntax**

```
class ClassName {
ClassName(parameters) {
// initialization code
}
}
```

- **Polymorphism (Method Overloading)**: Defining multiple methods with the same name but different parameters.
- **Syntax(e.g.)**

```
class  ClassName  {
int add(int a, int b) {
return a + b;
}

int add(int a, int b, int c) {
return a + b + c;
}
}
```

Post Lab:

1. Write a Java program to create a class and object
2. Write a Java program to demonstrate methods in a class.
3. Write a Java program to demonstrate polymorphism using method overloading
4. A college wants to store student details. Each student has a roll number and name.
   The system should:
   a. Initialize student details using constructors
   b. Display details using methods
   c. Calculate total marks using method overloading
5. An online shopping system needs to calculate the total bill.
   The program should:
   a. Initialize product details using constructors
   b. Calculate bill amount using methods
   c. Apply discount using method overloading

Answer

1.

Input

```
class phone{
String brand;
int price;
public static void main(String[] args) {
phone phone = new phone();
phone.brand = "Samsung";
```

```
phone.price = 25000;
System.out.println("Phone Brand: " + phone.brand);
System.out.println("Phone Price: " + phone.price);
}
}
```

Output

```
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\mpstme.student\Desktop\New folder> javac phone.java
PS C:\Users\mpstme.student\Desktop\New folder> java phone.java
Phone Brand: Samsung
Phone Price: 25000
PS C:\Users\mpstme.student\Desktop\New folder>
```

2.

Input

```
class main {
void display() {
System.out.println("This is a method inside a class");
}
public static void main(String[] args) {
main obj = new main();
obj.display(); // method call
}
}
```

Output

```
PS C:\Users\mpstme.student\Desktop\New folder> javac main.java
PS C:\Users\mpstme.student\Desktop\New folder> java main.java
This is a method inside a class
PS C:\Users\mpstme.student\Desktop\New folder>
```

3.

Input

```
class main {
void add(int a, int b) {
System.out.println("Sum: " + (a + b));
}
void add(int a, int b, int c) {
```

```
System.out.println("Sum: " + (a + b + c));
}
public static void main(String[] args) {
main obj = new main();
obj.add(105, 208);
obj.add(103, 210, 350);
}
}
```

Output

```
PS C:\Users\mpstme.student\Desktop\New folder> javac main.java
PS C:\Users\mpstme.student\Desktop\New folder> java main.java
Sum: 313
Sum: 663
```

4.

Input

```
class Student {
int rollNo;
String name;
Student(int r, String n) {
rollNo = r;
name = n;
}
void display() {
System.out.println("Roll No: " + rollNo);
System.out.println("Name: " + name);
}
int totalMarks(int m1, int m2) {
return m1 + m2;
}
int totalMarks(int m1, int m2, int m3) {
return m1 + m2 + m3;
}
public static void main(String[] args) {
Student s = new Student(147, "Saumya");
s.display();
System.out.println("Total Marks (2 subjects): " + s.totalMarks(45, 50));
System.out.println("Total Marks (3 subjects): " + s.totalMarks(40, 45, 50));
}
}
```

Output

```
PS C:\Users\mpstme.student\Desktop\New folder> javac Student.java
PS C:\Users\mpstme.student\Desktop\New folder> java Student.java
Roll No: 147
Name: Saumya
Total Marks (2 subjects): 95
Total Marks (3 subjects): 135
```

5.

Input

```
class Product {
String productName;

double price;
int quantity;
Product(String p, double pr, int q) {
productName = p;
price = pr;
quantity = q;
}
double calculateBill() {
return price * quantity;
}
double calculateBill(double discountPercent) {
double total = price * quantity;
return total - (total * discountPercent / 100);
}
public static void main(String[] args) {
Product p = new Product("Laptop", 50000, 1);
System.out.println("Total Bill: " + p.calculateBill());
System.out.println("Bill after 10% discount: " + p.calculateBill(10));
}
}
```

Output

```
PS C:\Users\mpstme.student\Desktop\New folder> javac Product.java
PS C:\Users\mpstme.student\Desktop\New folder> java Product.java
Total Bill: 50000.0
Bill after 10% discount: 45000.0
PS C:\Users\mpstme.student\Desktop\New folder>
```